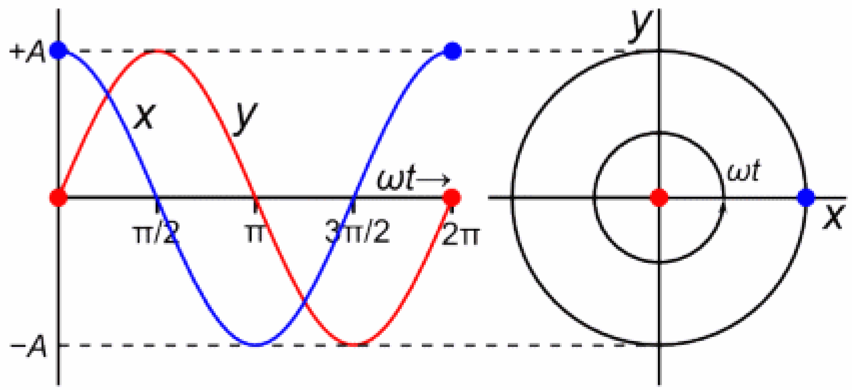
United International **University**

A Complete Guideline for Graphical Phase Change analysis in Simple Harmonic Motion



Phase, Phase shift, and Phase difference

TEXTBOOK SAYS: "Phase is a variable that depends on position and time."

$$\varphi(x,t) = kx \pm \omega t + \varphi$$

$$\varphi(x,t) = \left(\frac{2\pi}{\lambda}\right)x \pm 2\pi f t + \varphi$$

$$\theta = \omega t + \delta = \varphi$$

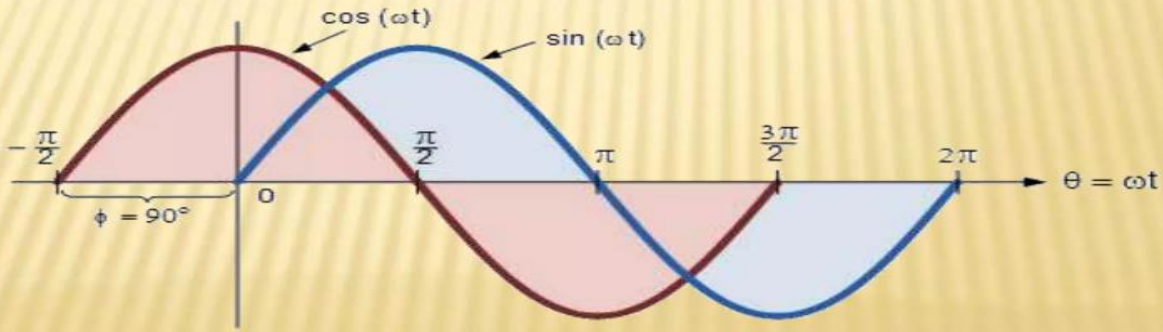
$$\theta = \varphi = \tan^{-1}\left(-\frac{v}{\omega x}\right)$$

$$\delta = \sin^{-1}\left(\frac{x_0}{A}\right) = \cos^{-1}\left(\frac{x_0}{A}\right) = \tan^{-1}\left(-\frac{v_0}{\omega_0 x_0}\right) = \sin^{-1}\left(-\frac{v_0}{\omega_0 A}\right)$$

The phase or phase angle of a sine signal is often noted with Φ or θ and measured in radians (rad) or degrees (°) and can vary between **-π and +π rad** or **-180° and +180°**.

The phase of periodic wave describes where the wave is in its cycle. δ is a phase constant.

The phase of a wave is dependant on position and time and allow us flexibility to describe the graph with a variety of different notations



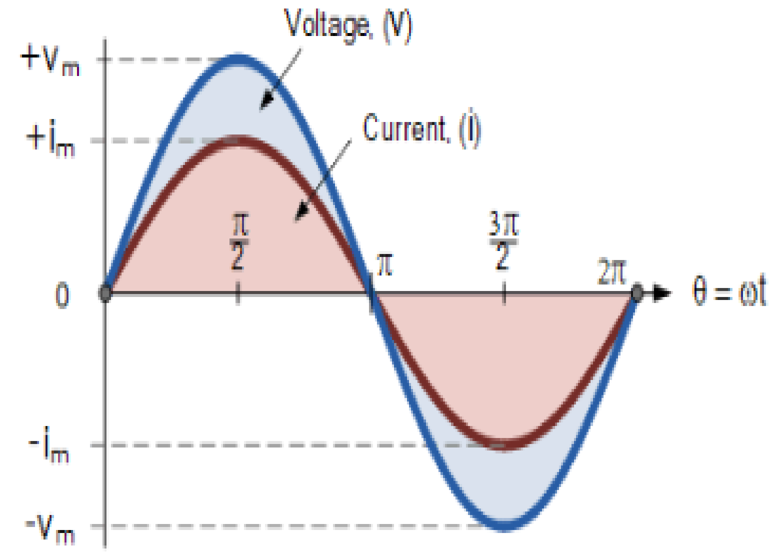
Phase, Phase shift, and Phase difference

$$\theta = \omega t + \delta = \varphi \qquad \therefore \theta = \varphi = \tan^{-1}(-\frac{v}{\omega x})$$

Phase is used to describe a specific location in a given cycle of a periodic wave.

It is time-dependent. So, we can say that the relative position of one point on a wave to another point on the same wave is known as Phase.

Phases are of two types- in phase and out of phase.



In Phase, waves are the waves which are 0∘ or 360∘ apart.

The resulting amplitude of in-phase waves is twice the original wave.

Out-of-phase waves are the waves that are 180∘apart.

The resulting amplitude of the completely out-of-phase wave is zero.

Phase, Phase shift, and Phase difference

- Phase difference is used to describe the phase position of one wave relative to another.
- ***At a glance, the difference of phase position of the waves is called phase difference.***

Usually expressed by δ or Δ in angle.

- **PHILIP SAYS:** The phase difference, $\Delta\varphi$, can be thought of as the value between two different phases.

$$\Delta\varphi = \varphi_2 - \varphi_1$$

- If we plug in the two phases we identified earlier, we can determine the phase constant.

$$\Delta\varphi = (kx_0 + \omega t_2 + \varphi) - (kx_0 + \omega t_1 + \varphi)$$
$$= \omega t_2 + \varphi - \omega t_1 + \varphi$$

This represents an instance where two sinusoidal waves with different initial displacements are observed at one position over time, and we would like to know what the difference between their phases are.

Phase, Phase shift, and Phase difference

On a graph, the phase of an AC signal represents the initial state of its related sine function at the origin of time :

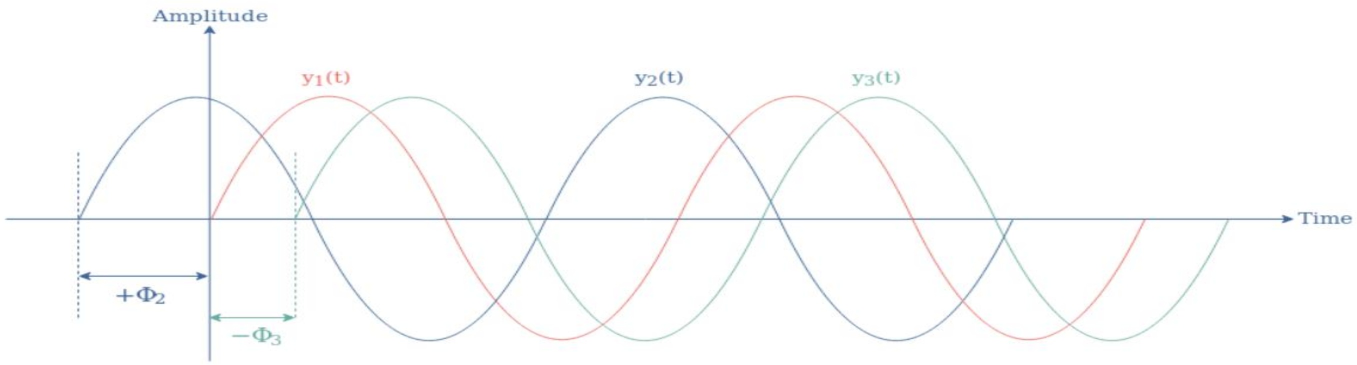
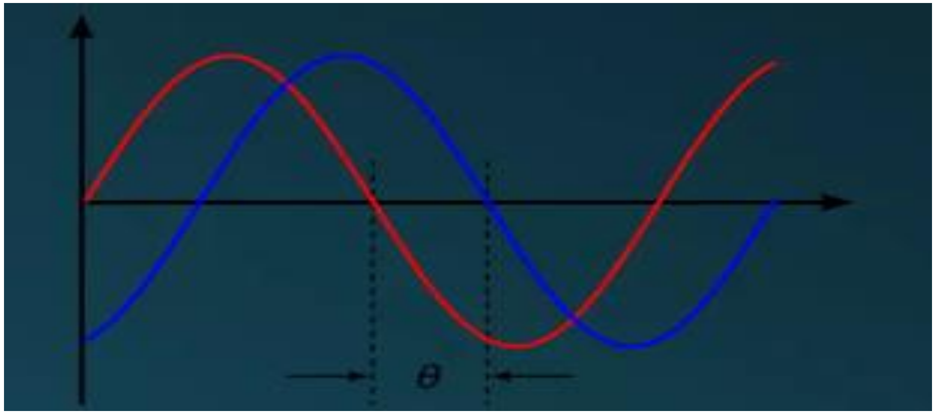


fig 1: Illustration of three sine waveforms with different phases

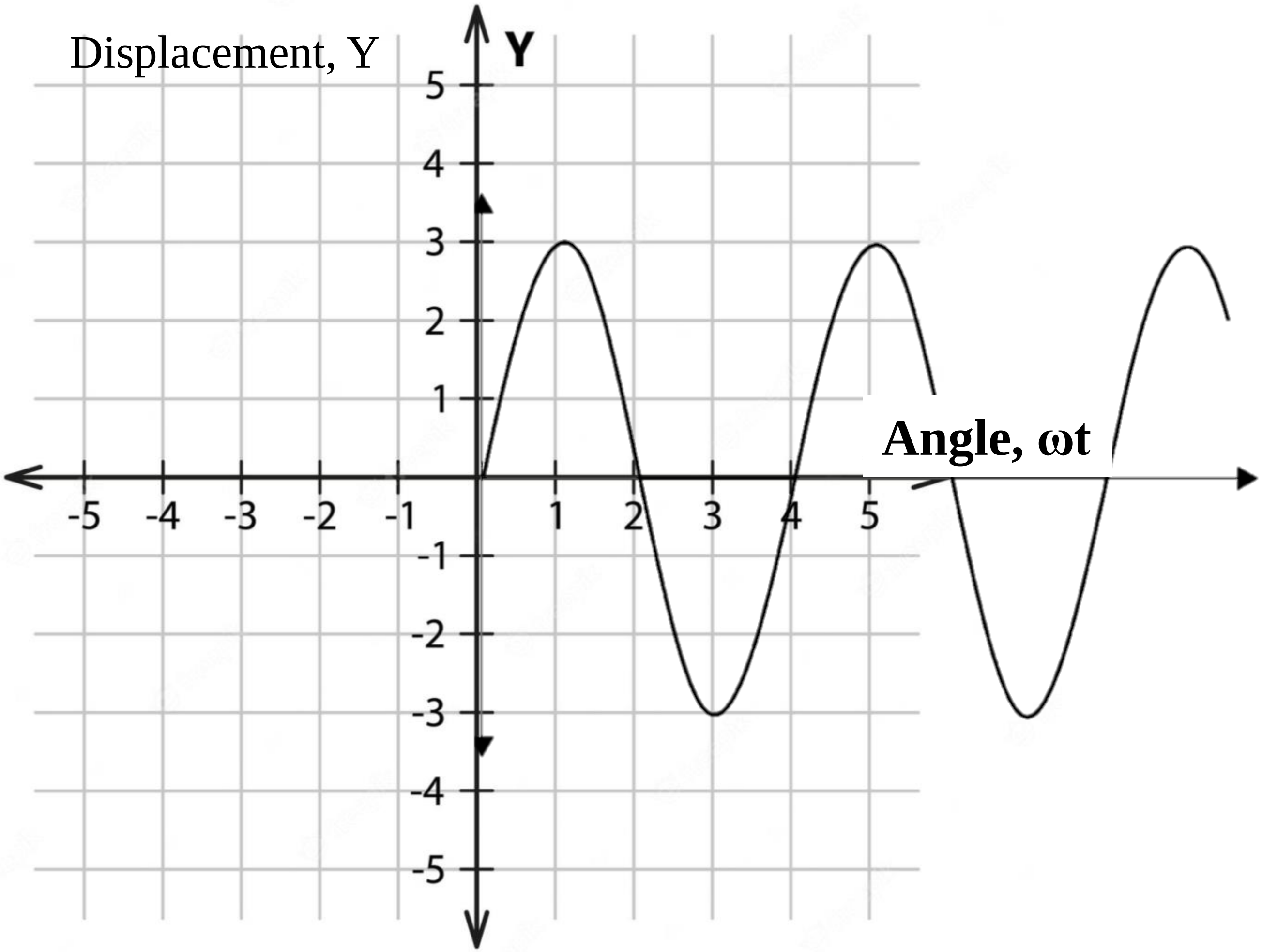
The phase Φ of a signal can be of three different nature and dictates the position of the waveform around the vertical axis :

1.Equal to 0 (° or rad) such as for the signal $y_1(t)$ which act as a reference signal
2.Be positive such as for the signal $y_2(t)$
3.Be negative such as for the signal $y_3(t)$



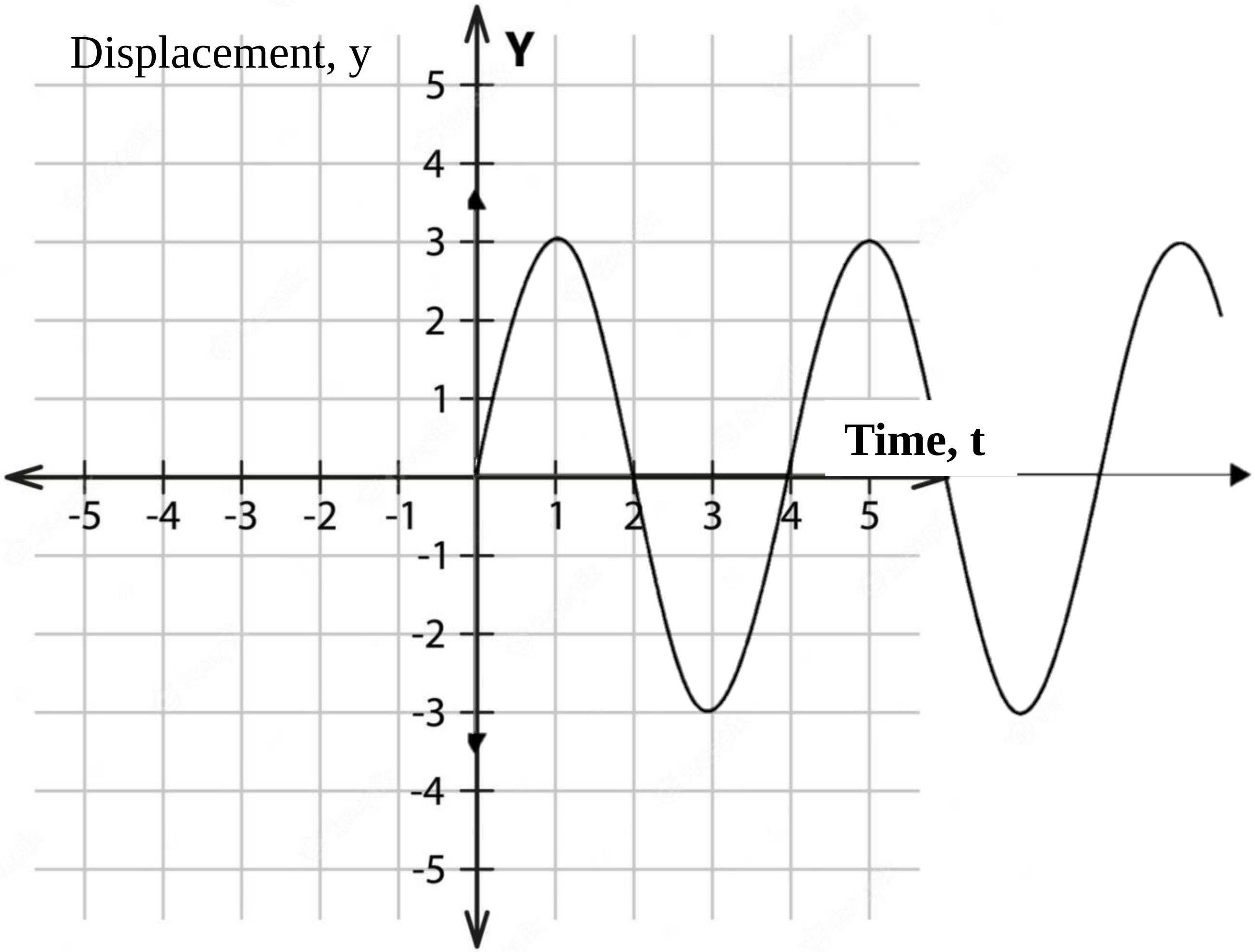
Phase, Phase shift, and Phase difference

Let us consider the displacement of a simple harmonic oscillator is, **$y = A \sin\omega t$**
Let's see the graph can be represented as



Phase, Phase shift, and Phase difference

We can also plot this **y = A sinωt** graph for y vs. t and the graph will be like the figure below



Phase, Phase shift, and Phase difference

Now if we consider 2nd particle with $\mathbf{y = A \sin(\omega t+\varphi)}$

Displacement, $\mathbf{y = A \sin(\omega t+\varphi)}$

Now if we want to calculate the value of wt for which the value of **y= 0**

For $\mathbf{y = 0}$, $\mathbf{\sin(\omega t+\varphi) = 0}$

$$\mathbf{(\omega t+\varphi) = 0}$$

$$\mathbf{\omega t = -\varphi \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(1)}$$

So the wave pattern of the particle will originates from an angle, $\mathbf{\omega t = -\varphi}$

if we consider four cases of displacements for graphical representation

(i) $\mathbf{y = A \sin(\omega t+ \frac{\pi}{4})}$

(ii) $\mathbf{y = A \sin(\omega t - \frac{\pi}{4})}$

(iii) $\mathbf{y = A \sin(\omega t + \frac{\pi}{2})}$

(iv) $\mathbf{y = A \sin(\omega t + \frac{\pi}{3})}$

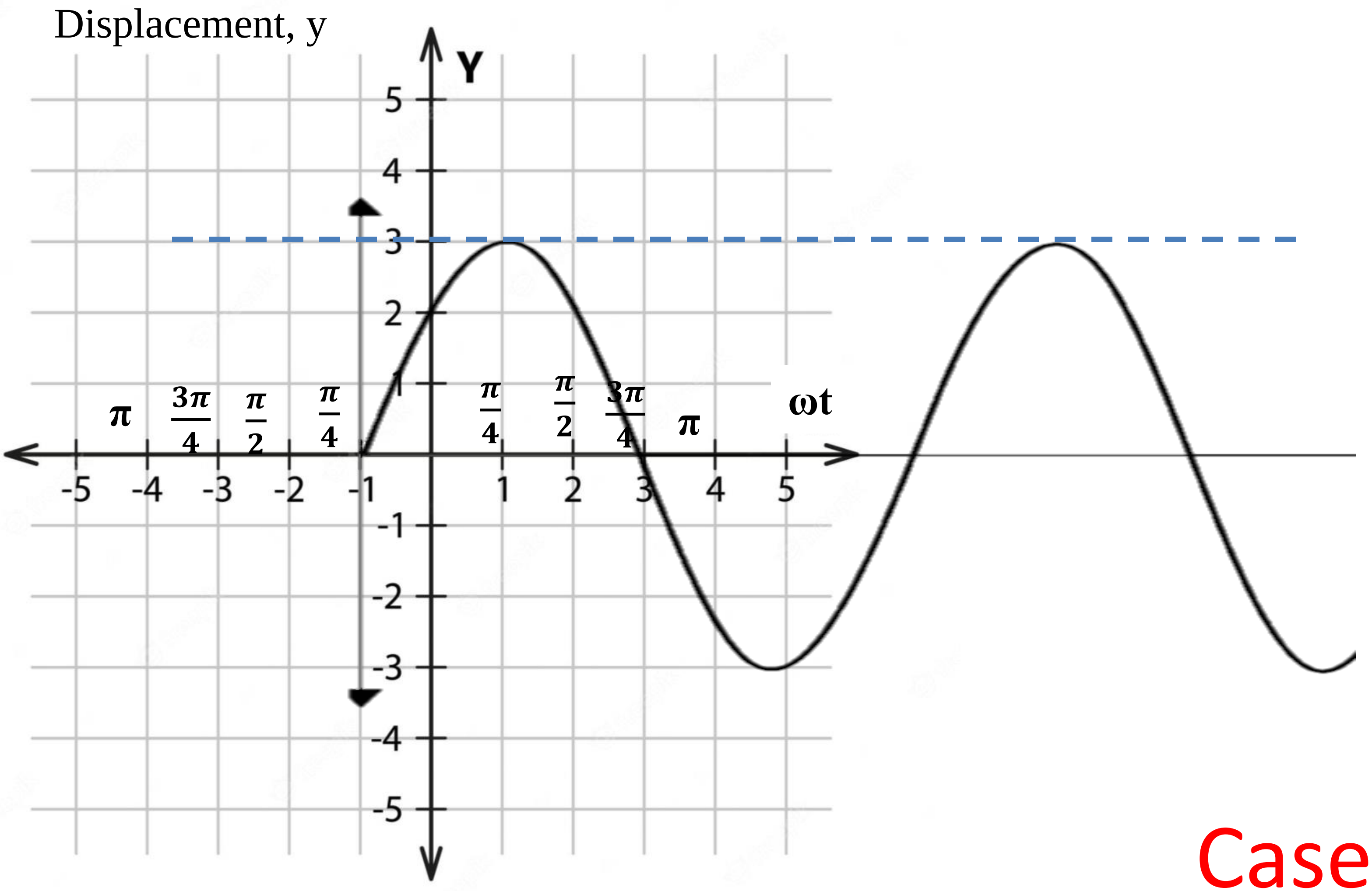
Please note that, you must plot the graph from $\omega t = 0$ position. We have only provided the negative value of ωt for primary understanding and practice purpose. But you must start the graph from wt = 0 when you understand the patterns.

Phase, Phase shift, and Phase difference

Case (i)

$y = A\sin(\omega t + \frac{\pi}{4})$

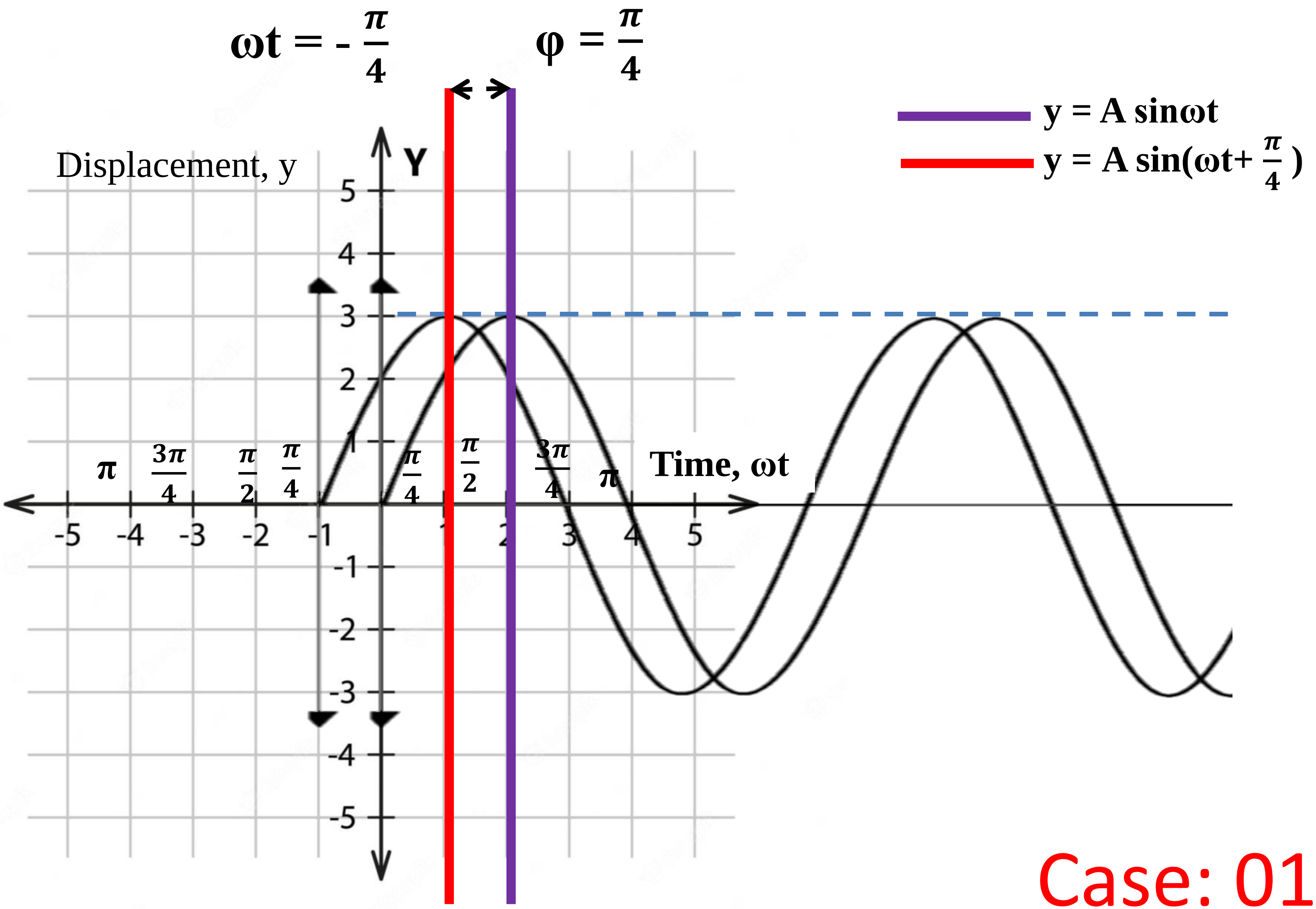
From Eq^n (1), the graph will start from $\boldsymbol{\omega t = -\frac{\pi}{4}}$



Case: 01

Phase, Phase shift, and Phase difference

Displacement y vs. Angle ωt

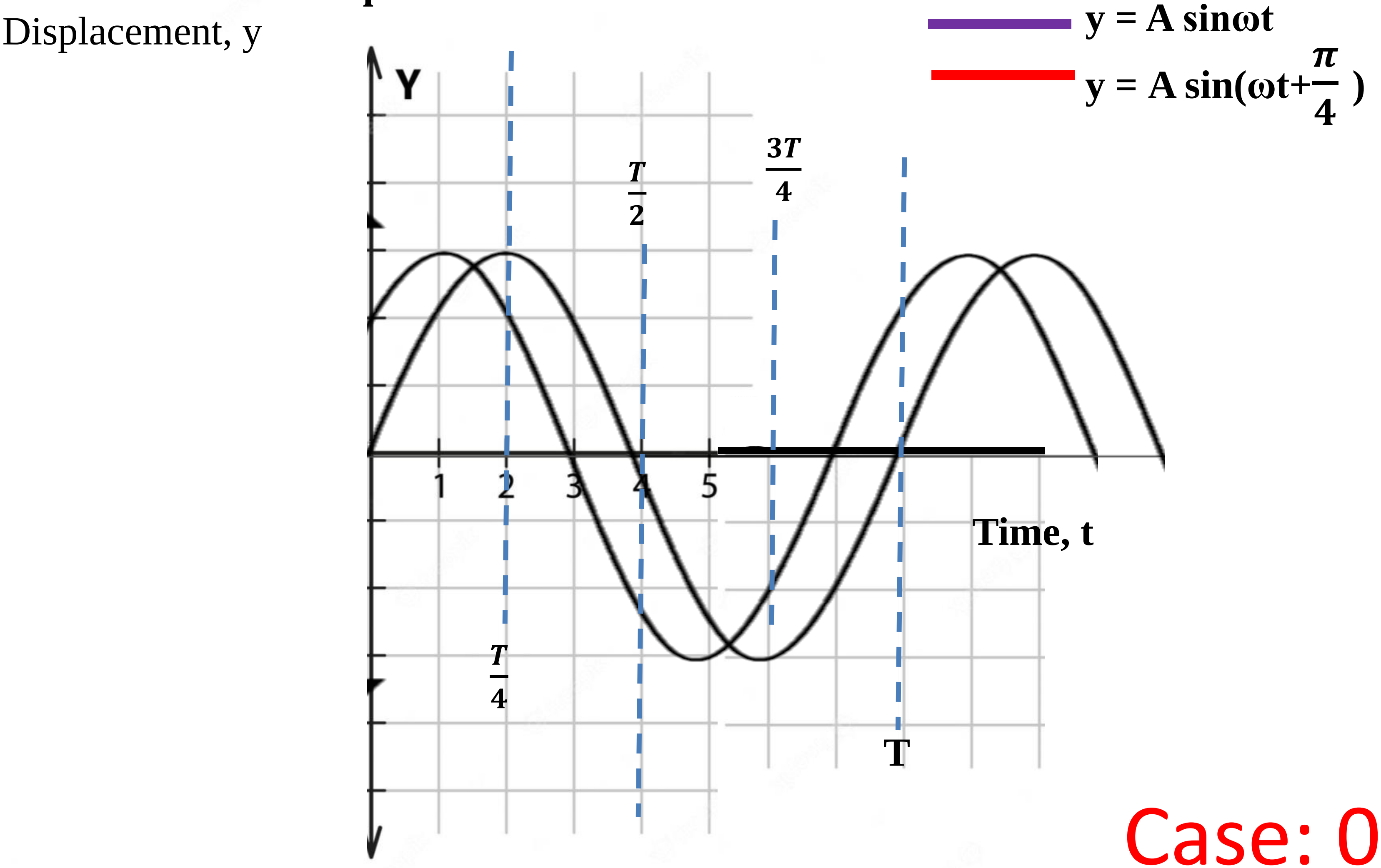


Case: 01

Phase, Phase shift, and Phase difference

Displacement y vs. Time t

$\omega t = -\frac{\pi}{4}$ $\varphi = \frac{\pi}{4}$



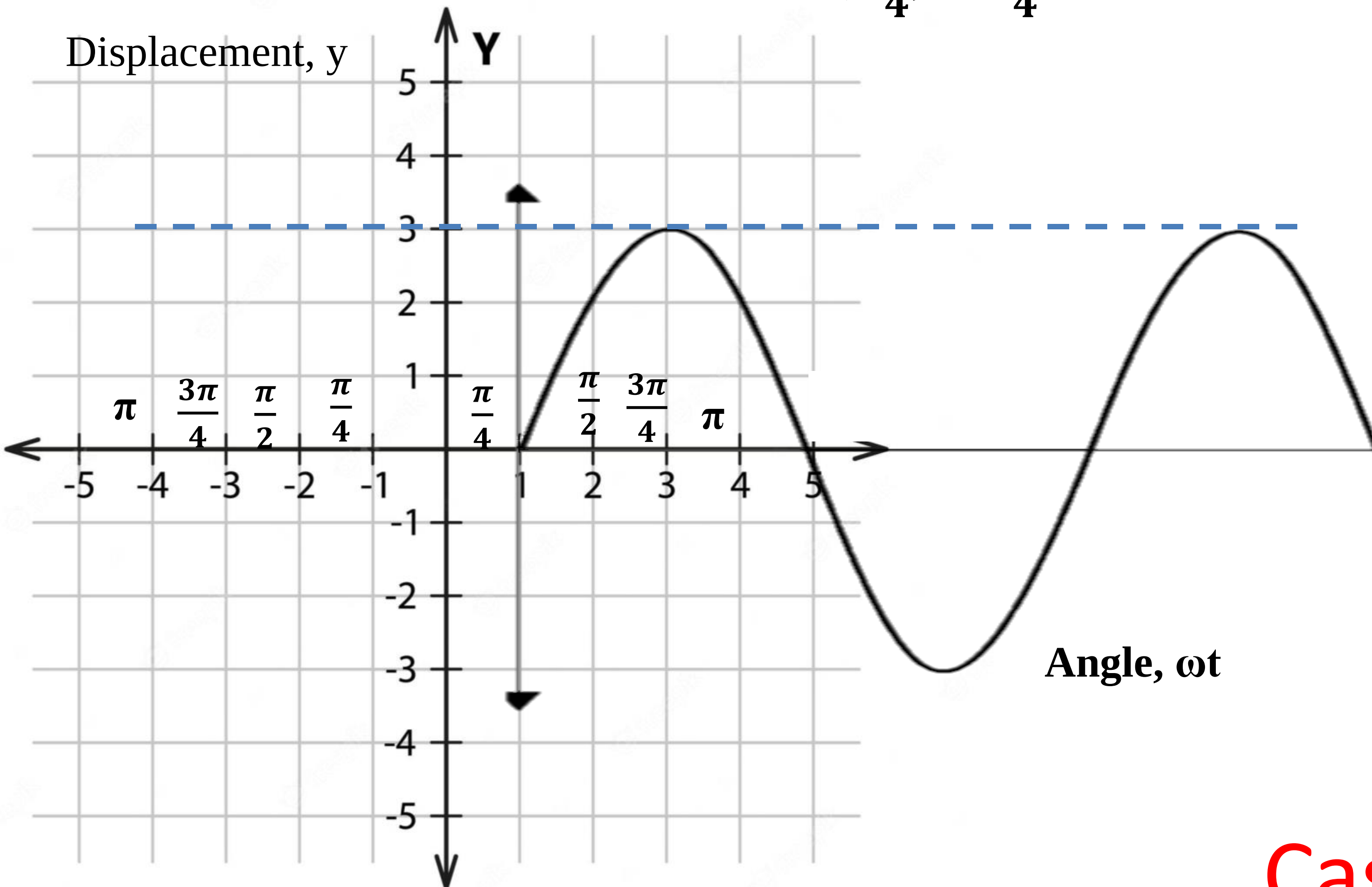
Case: 01

Phase, Phase shift, and Phase difference

Case (ii)

$$y = A\sin(\omega t - \frac{\pi}{4})$$

From Eq^{n} (1), the graph will start from $\omega t = -(-\frac{\pi}{4}) = \frac{\pi}{4}$

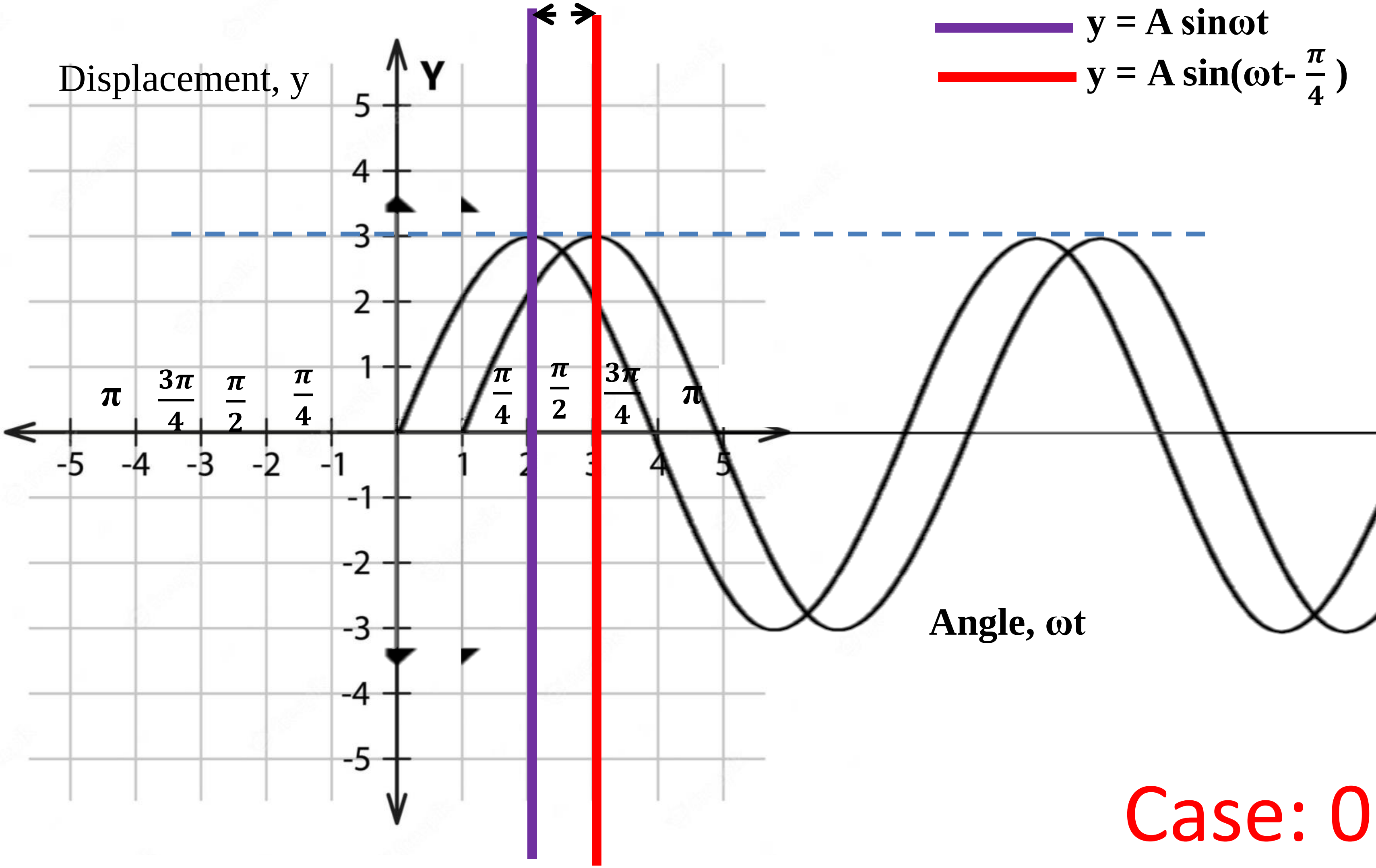


Case: 02

Phase, Phase shift, and Phase difference

Displacement y vs. Angle ωt

$\omega t = \frac{\pi}{4}$ $\varphi = -\frac{\pi}{4}$

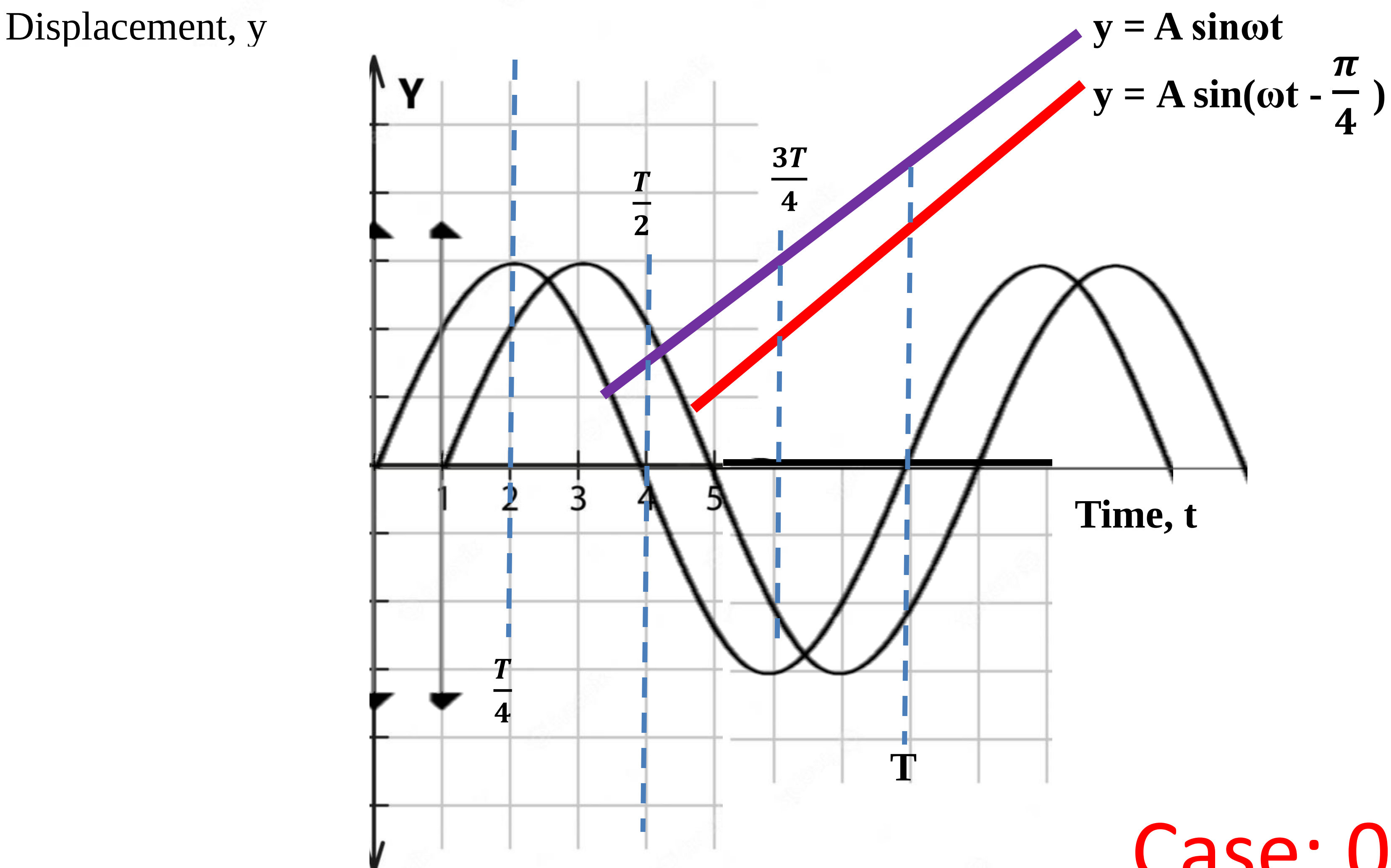


Case: 02

Phase, Phase shift, and Phase difference

Displacement y vs. Time t

$\omega t = \frac{\pi}{4}$ $\varphi = -\frac{\pi}{4}$



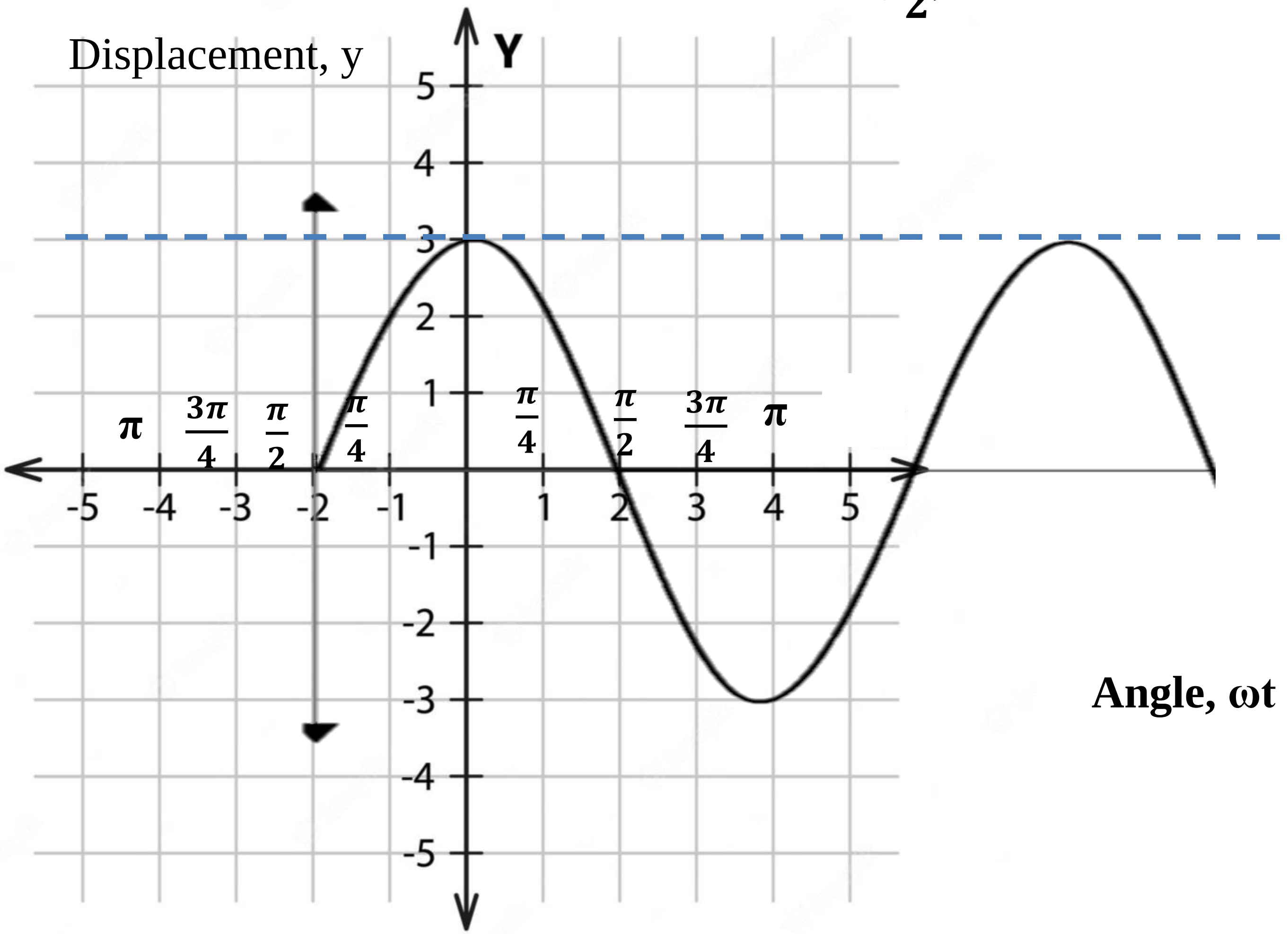
Case: 02

Phase, Phase shift, and Phase difference

Case (iii)

$y = A \sin(\omega t + \frac{\pi}{2})$

From Eq^n (1), the graph will start from $\boldsymbol{\omega t = -(\frac{\pi}{2})}$

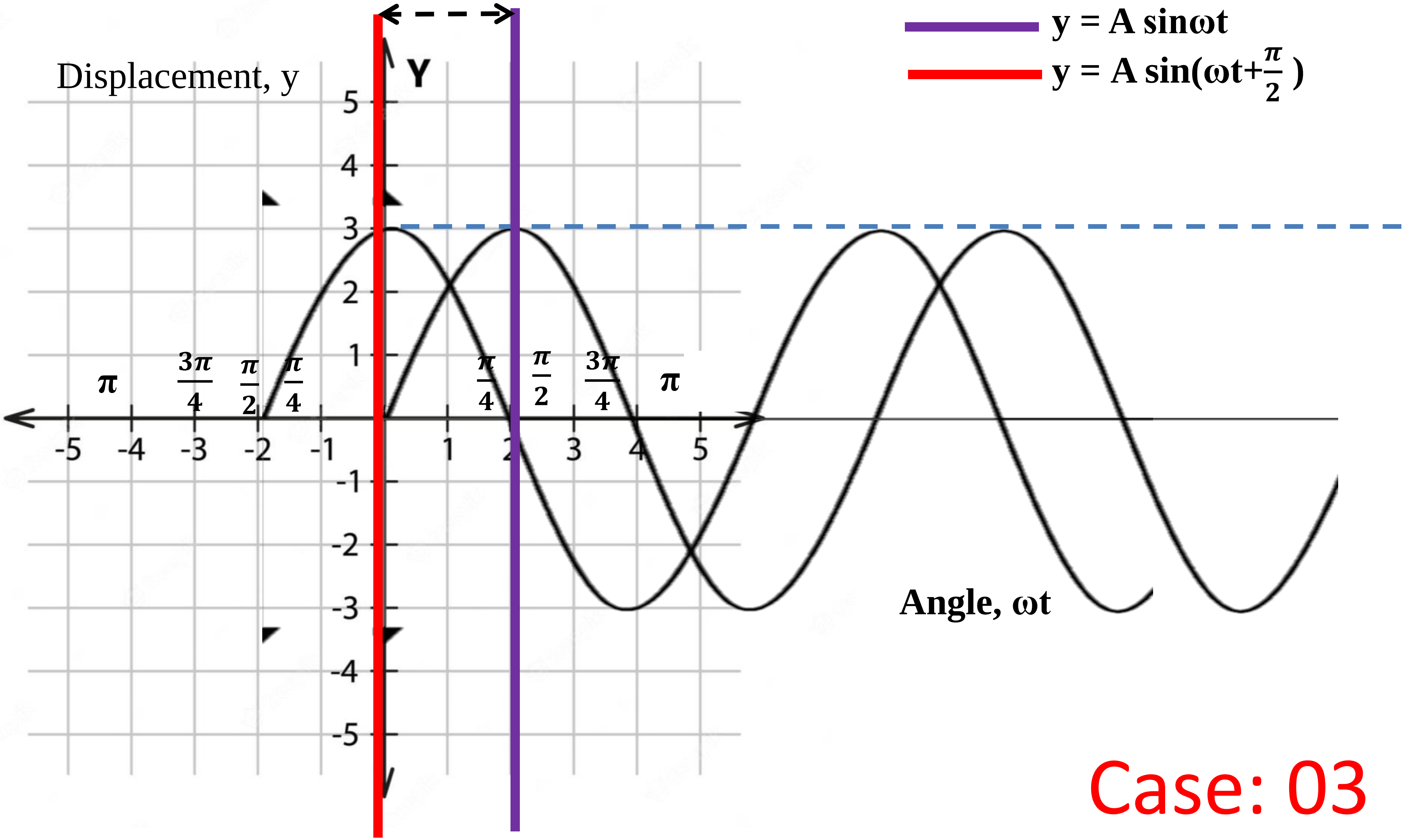


Case: 03

Phase, Phase shift, and Phase difference

Displacement y vs. Angle ωt

$\omega t = -\frac{\pi}{2}$ $\varphi = \frac{\pi}{2}$

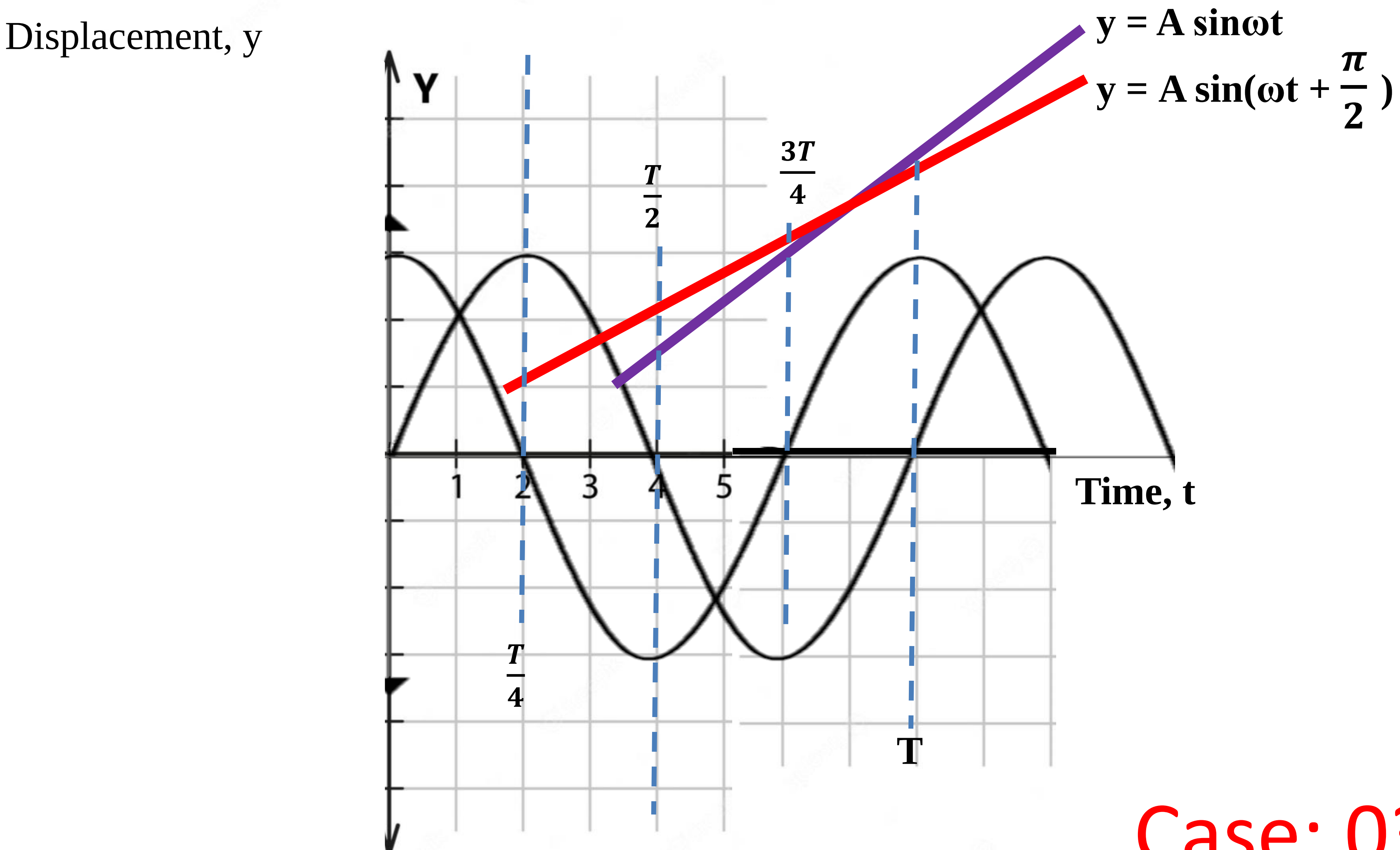


Case: 03

Phase, Phase shift, and Phase difference

Displacement y vs. Time t

$\omega t = -\frac{\pi}{2}$ $\varphi = \frac{\pi}{2}$



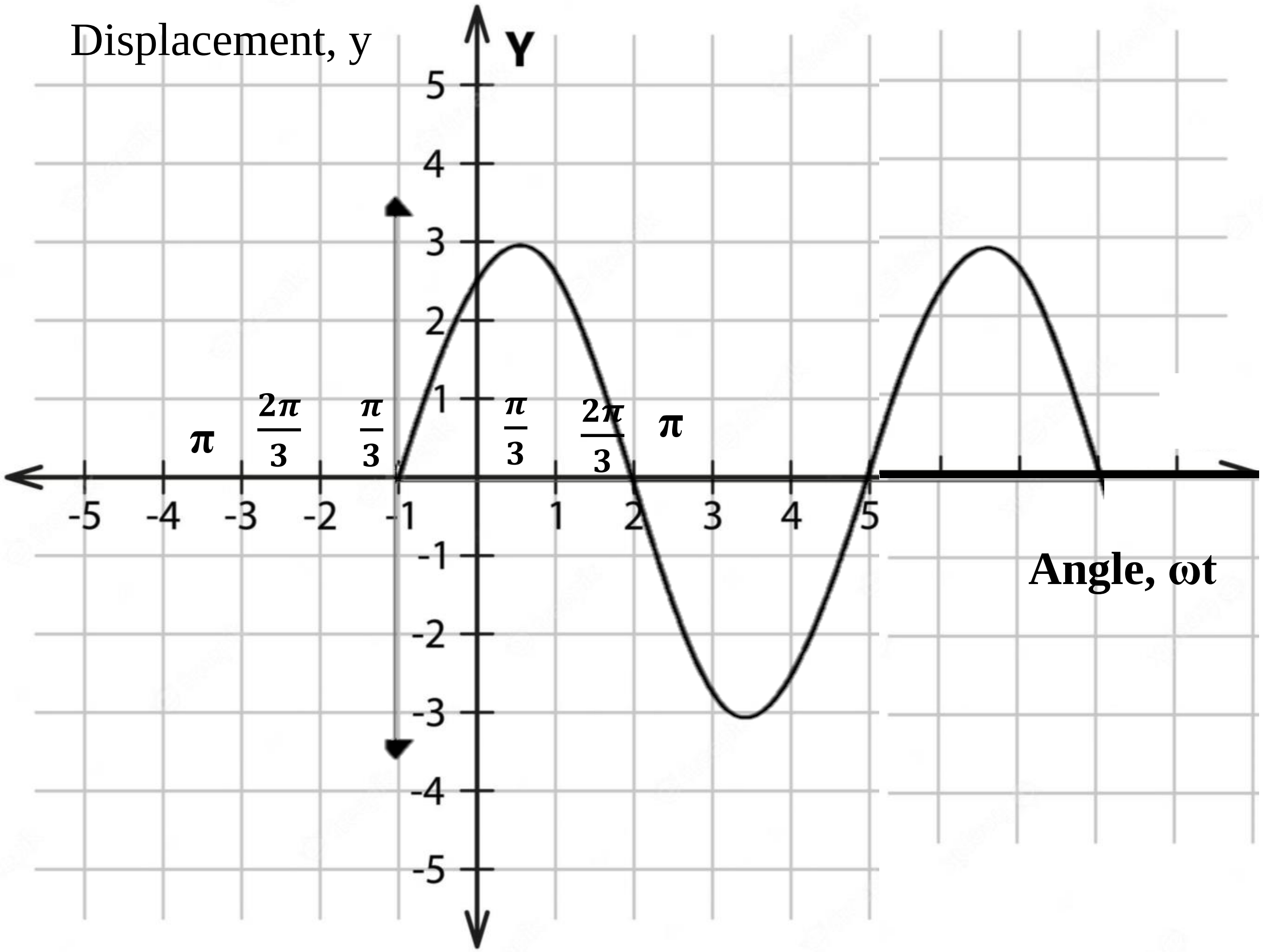
Case: 03

Phase, Phase shift, and Phase difference

Case (iv):

Now if we consider the particle with $\mathbf{y = A \sin(\omega t + \frac{\pi}{3})}$

graph for **y** vs. **ωt** and the graph will be like the figure below

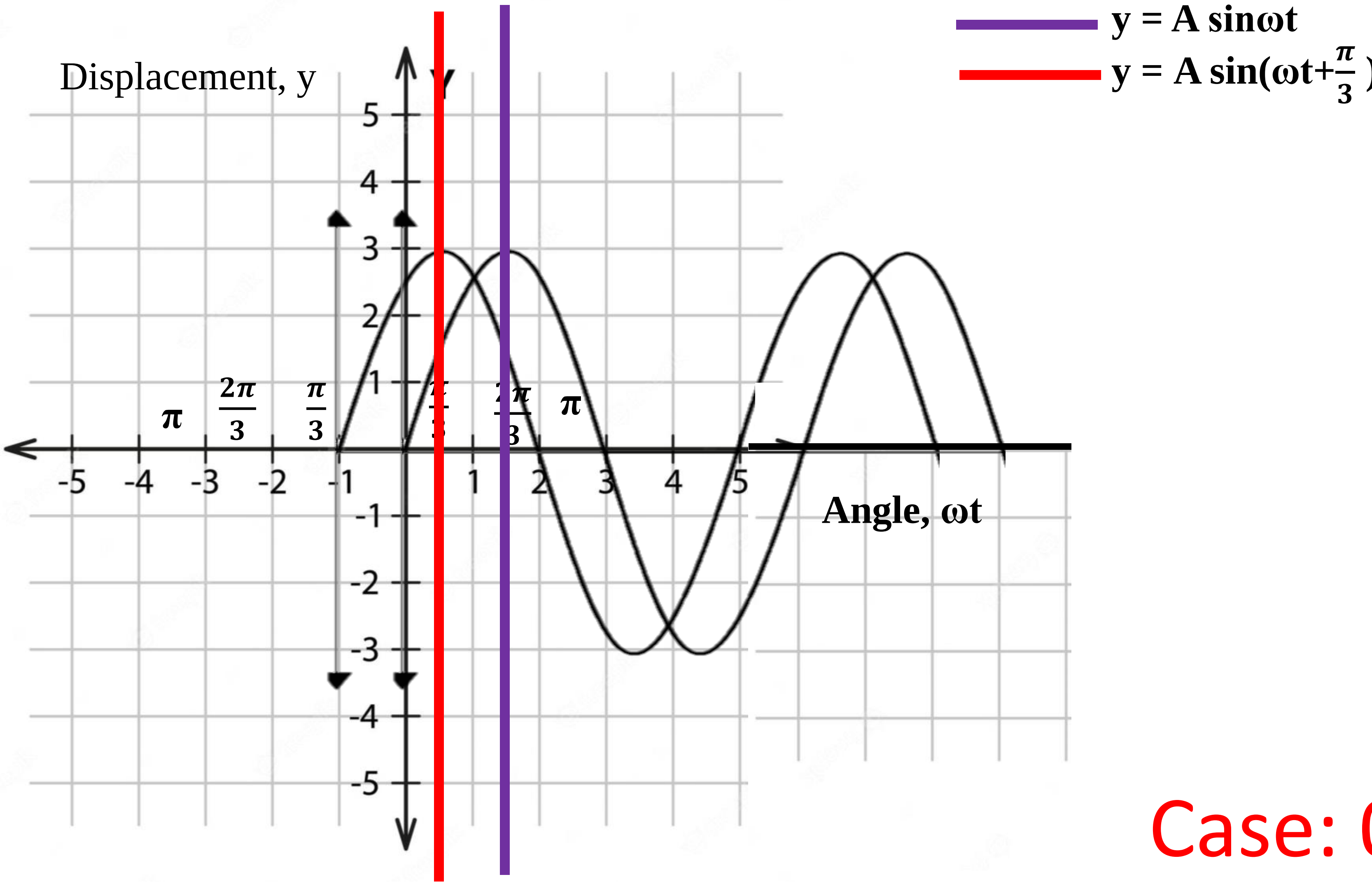


Case: 04

Phase, Phase shift, and Phase difference

Displacement y vs. Angle ωt

$\omega t = -\frac{\pi}{3}$ $\varphi = \frac{\pi}{3}$

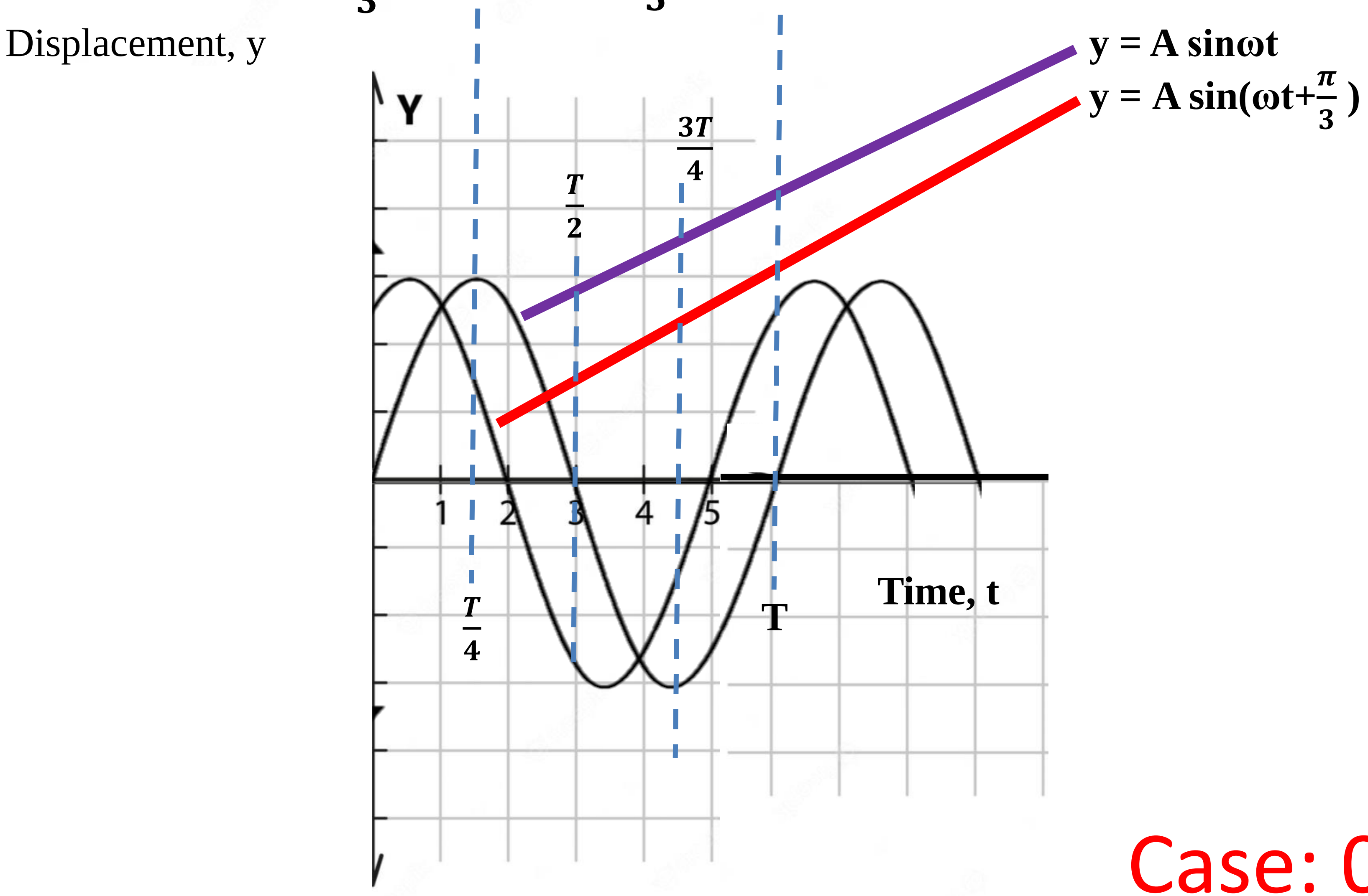


Case: 04

Phase, Phase shift, and Phase difference

Displacement y vs. time t

$\omega t = -\frac{\pi}{3}$ $\varphi = \frac{\pi}{3}$



Case: 04

Phase, Phase shift, and Phase difference

Some Practice Problems

(i) $y = A \sin(\omega t - \frac{\pi}{3})$

(ii) $y = A \sin(\omega t - \frac{\pi}{2})$

(iii) $y = A \sin(\omega t + \pi)$

(iv) $y = A \sin(\omega t - \pi)$

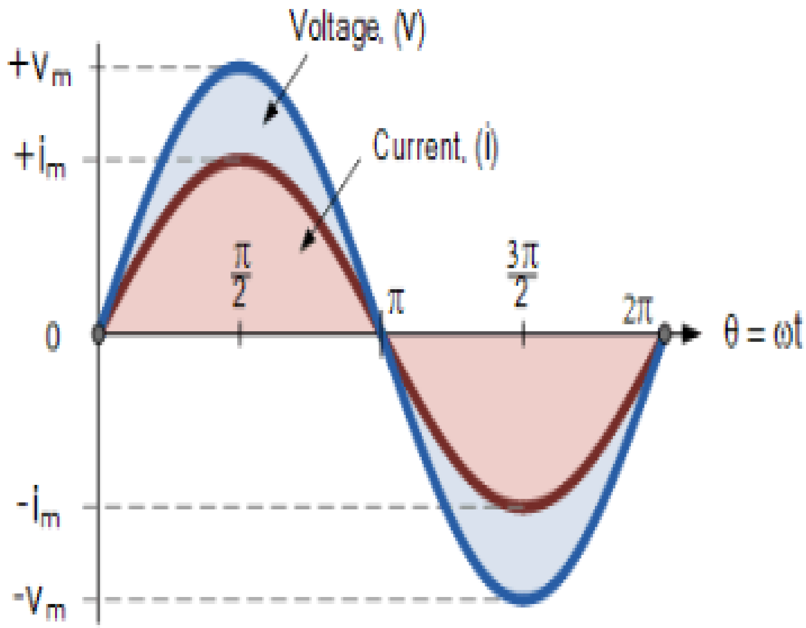
(v) $y = A \sin(\omega t + \frac{3\pi}{4})$

(vi) $y = A \sin(\omega t - \frac{3\pi}{4})$

(vii) $y = A \sin(\omega t + \frac{3\pi}{2})$

(viii) $y = A \sin(\omega t + 2\pi)$

Phase, Phase shift, and Phase difference

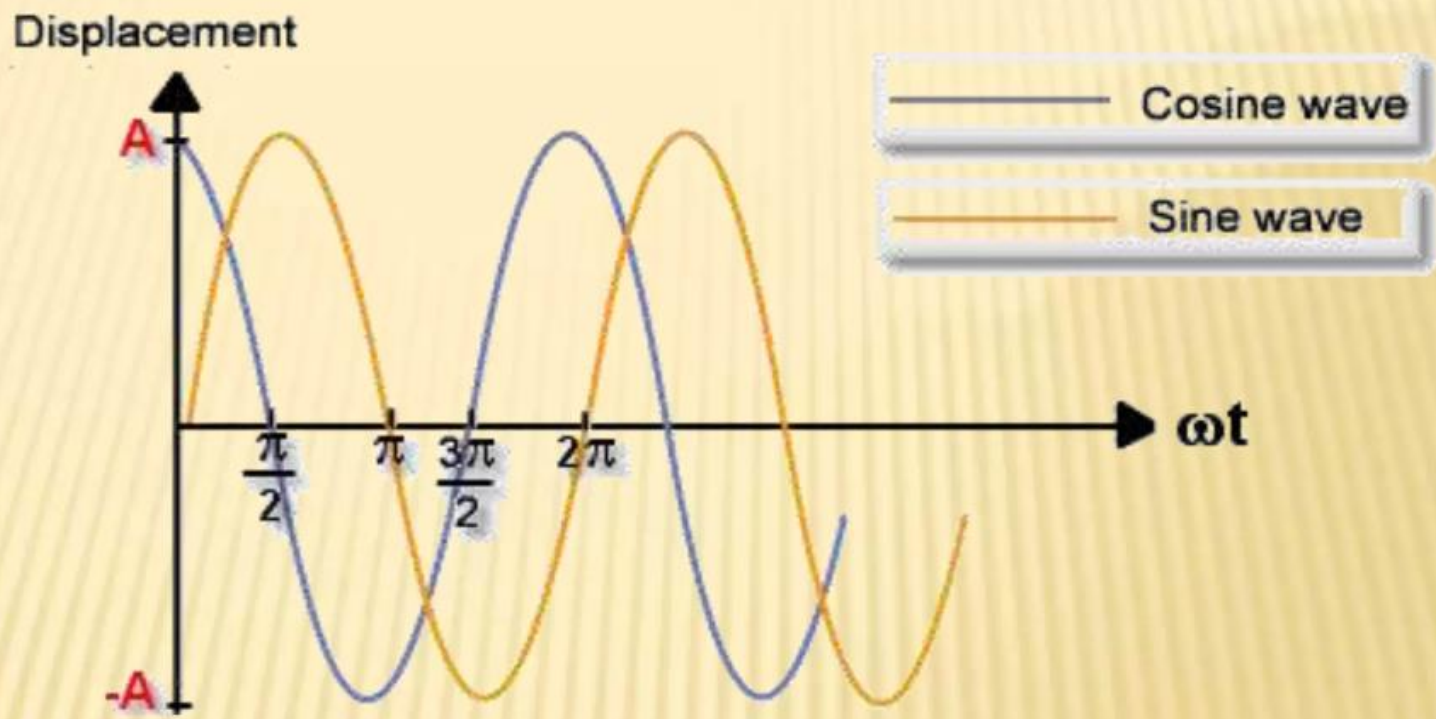


Phase Difference and Phase Shift

Phase Difference is used to describe the difference in degrees or radians when two or more alternating quantities reach their maximum or zero values

Phase, Phase shift, and Phase difference

Phase Difference:



- Here, we can take the cosine wave to be our reference. The sine waves can be described as being ahead of the cosine wave by π/2 . This means that there is a **phase difference** of π/2 between the two waves. The phase difference is the difference between the phases at two points at the same time t. Oscillations can have phase differences of any multiple of π. However, if they have a phase difference of either 0 or 2π they are said to be **in phase**.

Phase, Phase shift, and Phase difference

Phase Difference Equation

$$A_{(t)} = A_{max} \times \sin(\omega t \pm \Phi)$$

Where:

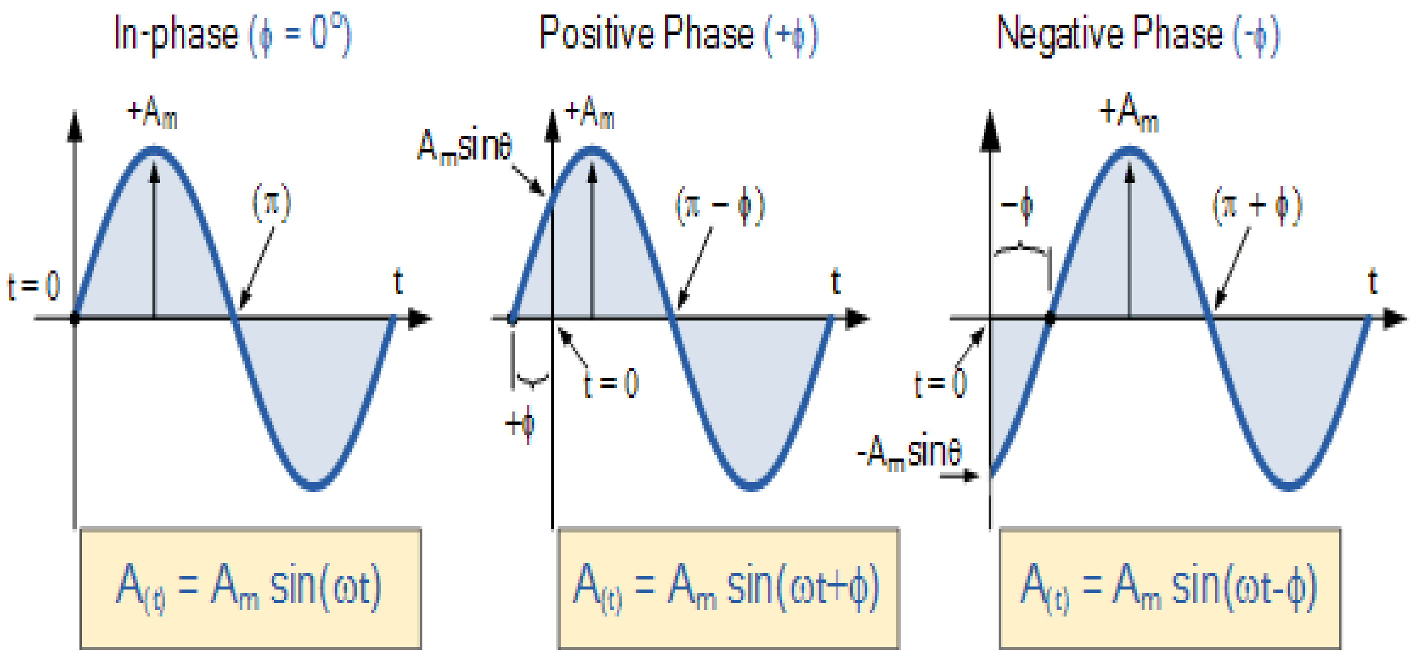
A_m – is the amplitude of the waveform.

ωt – is the angular frequency of the waveform in radian/sec.

Φ (phi) – is the phase angle in degrees or radians that the waveform has shifted either left or right from the reference point.

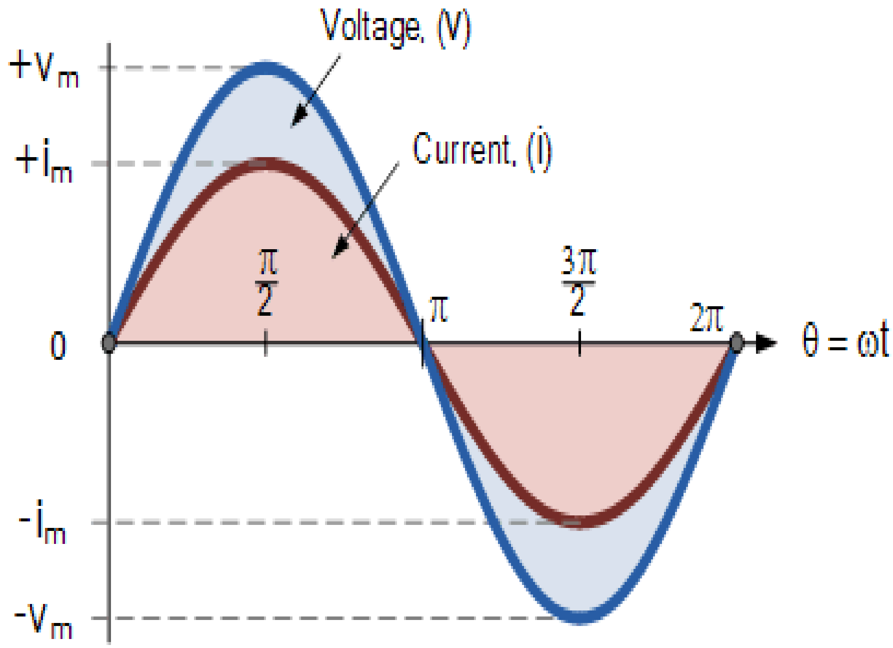
Phase, Phase shift, and Phase difference

Phase Relationship of a Sinusoidal Waveform



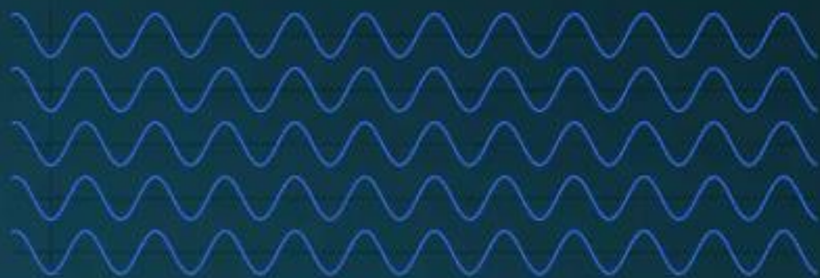
Phase, Phase shift, and Phase difference

Two Sinusoidal Waveforms – "in-phase"



Phase, Phase shift, and Phase difference

- After every race, Phillip graphs out the wave that he has ridden on. Sometimes, these waves are **in-phase:**

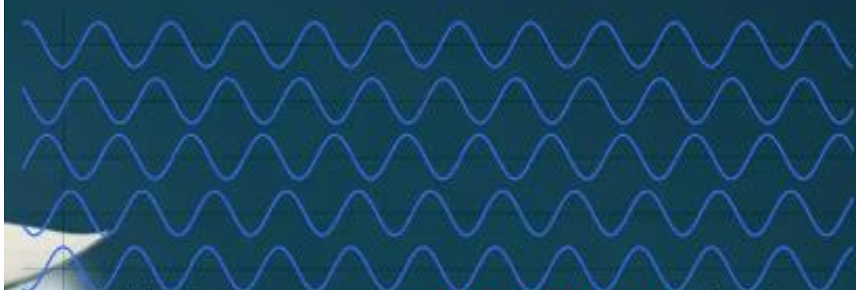


$\varphi_2 - \varphi_1 = n\,\lambda$

The phase difference of such waves are an even multiple of pi.

(This is known as constructive interference)

Other instances, the waves are **out of phase.**



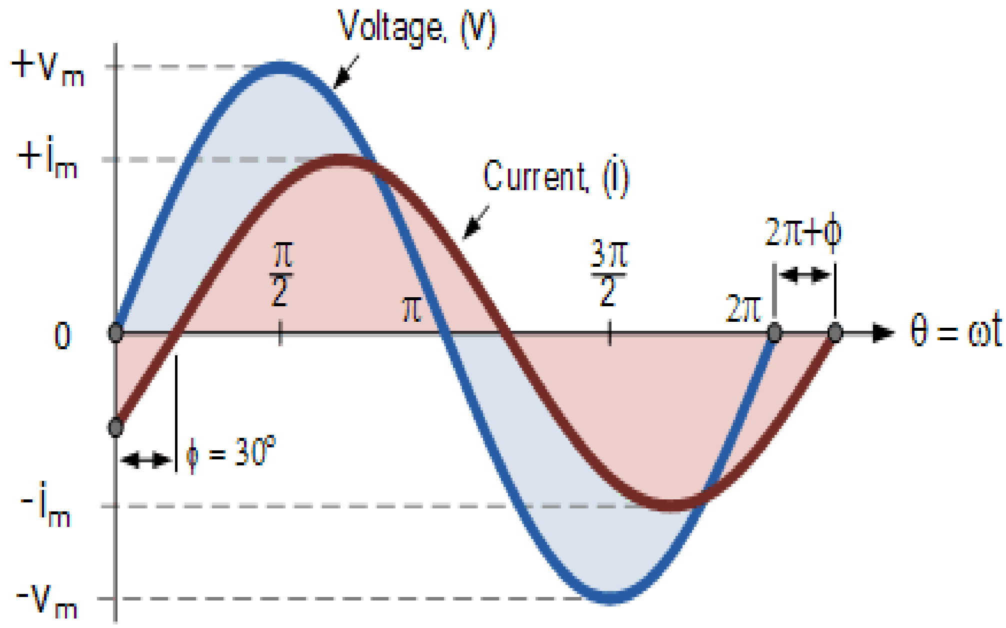
$\varphi_2 - \varphi_1 = (n + 0.5)\,\lambda$

The phase difference of such waves are an odd multiple of pi.

(This is known as destructive interference)

Phase, Phase shift, and Phase difference

Phase Difference of a Sinusoidal Waveform



Phase, Phase shift, and Phase difference

As the two waveforms are no longer "in-phase", they must therefore be "out-of-phase" by an amount determined by phi, Φ and in our example this is 30^o. So we can say that the two waveforms are now 30^o out-of phase. The current waveform can also be said to be "lagging" behind the voltage waveform by the phase angle, Φ. Then in our example above the two waveforms have a **Lagging Phase Difference** so the expression for both the voltage and current above will be given as.

$$\text{Voltage}, (v_t) = V_m \sin \omega t$$

$$\text{Current}, (i_t) = I_m \sin(\omega t - \phi)$$

Where current, i "lags" voltage, v by phase angle Φ

Phase, Phase shift, and Phase difference

Likewise, if the current, i has a positive value and crosses the reference axis reaching its maximum peak and zero values at some time before the voltage, v then the current waveform will be "leading" the voltage by some phase angle. Then the two waveforms ar said to have a **Leading Phase Difference** and the expression for both the voltage and the current will be.

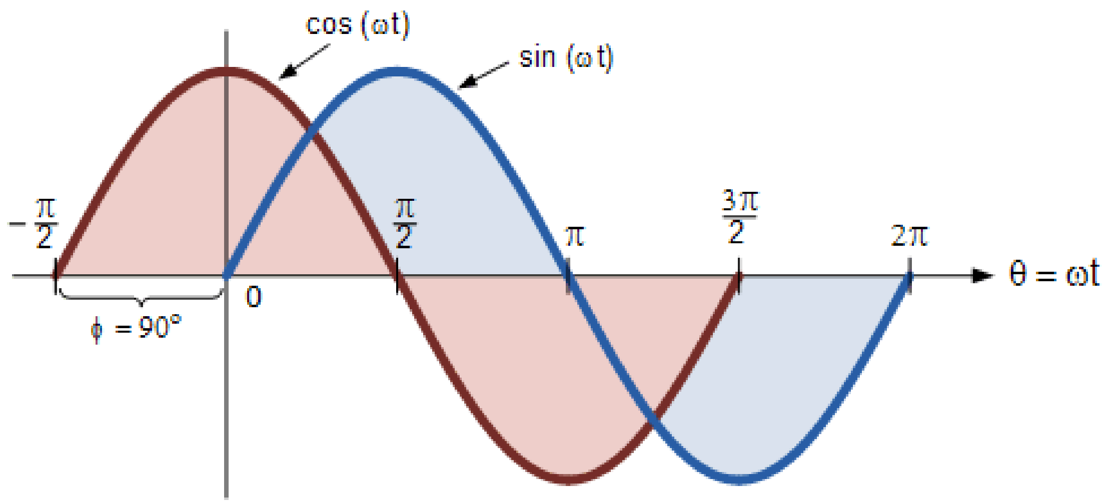
$$\text{Voltage}, (v_t) = V_m \sin \omega t$$

$$\text{Current}, (i_t) = I_m \sin(\omega t + \phi)$$

Where current, i "leads" the voltage v by phase angle Φ

Phase, Phase shift, and Phase difference

Difference between a Sine wave and a Cosine wave



Alternatively, we can also say that a sine wave is a cosine wave that has been shifted in the other direction by -90°. Either way when dealing with sine waves or cosine waves with an angle the following rules will always apply.

Phase, Phase shift, and Phase difference

Thank You

The END